

JKBOSE 2026

CLASS 11TH

Bhaicharra
ON
TOP.

#XUV 

ORGANIC CHEMISTRY:

SOME BASIC PRINCIPLES
AND TECHNIQUES

LECTURE-11

CHEMISTRY

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Topics To Covered

1

Concept of Electrophile and Nucleophile

2

Covalent Bond Fission and Intermediates

REVISION.

$\bar{e}$ -displacements effects.

Electromeric effect

a) $+\epsilon$

b) $-\epsilon$

Temporary

$\pi\bar{e}$.

Inductive

a) $+I$

b) $-I$

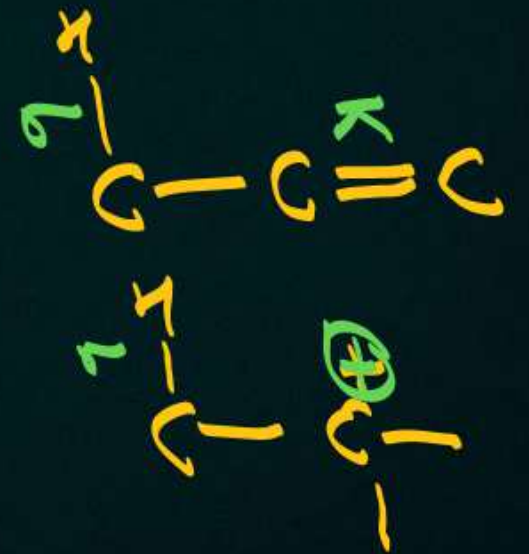
$\sigma\bar{e}$ displacements

Resonance

a) $+R$ (lone pair)

b) $-R$ (MB).

Hyperconjugation



CONCEPT OF ELECTROPHILE AND NUCLEOPHILE $\rightarrow$



1) ELECTROPHILE $\rightarrow$ Electrophiles are electron loving species. Their attraction for electron is due to the presence of e^- deficient atom in them. Electrophiles may be positively charged or Neutral species.

a) Positively charged electrophiles $\rightarrow H^+, Cl^+, I^+, NO^+, R^+, NO_2^+$ etc.

b) Neutral electrophiles $\rightarrow AlCl_3, BF_3, SnCl_4, FeCl_3, \dot{R}, SO_3$ etc.

• Electrophiles behave as Lewis acids (e^- pair acceptor).

Note $\rightarrow NH_4^+, Na^+, Ca^{+2}$ etc can't act as electrophile because their outermost shell is full.

E^+

2) NUCLEOPHILE $\rightarrow$ are nucleus (+ve charge) loving species. These are e^- rich species containing at least one lone pair of e^- . It can be negatively charged or neutral.

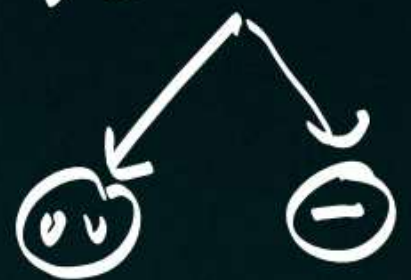
a) Negatively charged $\rightarrow H^-, Cl^-, I^-, Br^-, R^-, OH^-, NO_2^-, CN^-$ etc.

b) Neutral nucleophiles $\rightarrow H_2O:, NH_3:, R-OH, R\ddot{O}H, R-\ddot{N}H_2$ etc.

• Nucleophiles act as Lewis base (e^- pair donor).

Nu^-

Q. Which of the following can act as nucleophile $\Rightarrow$ (nu). nucleus loving $\rightarrow$ $\bar{e}$ rich



$\bar{e}$ deficient
1st

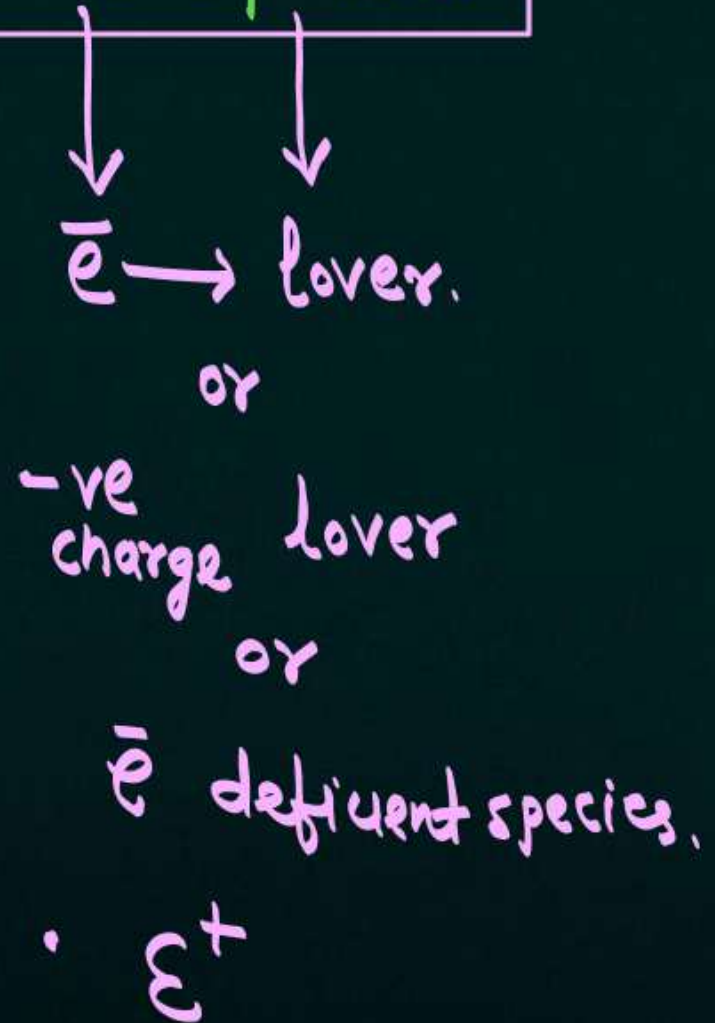


6 $\bar{e}$

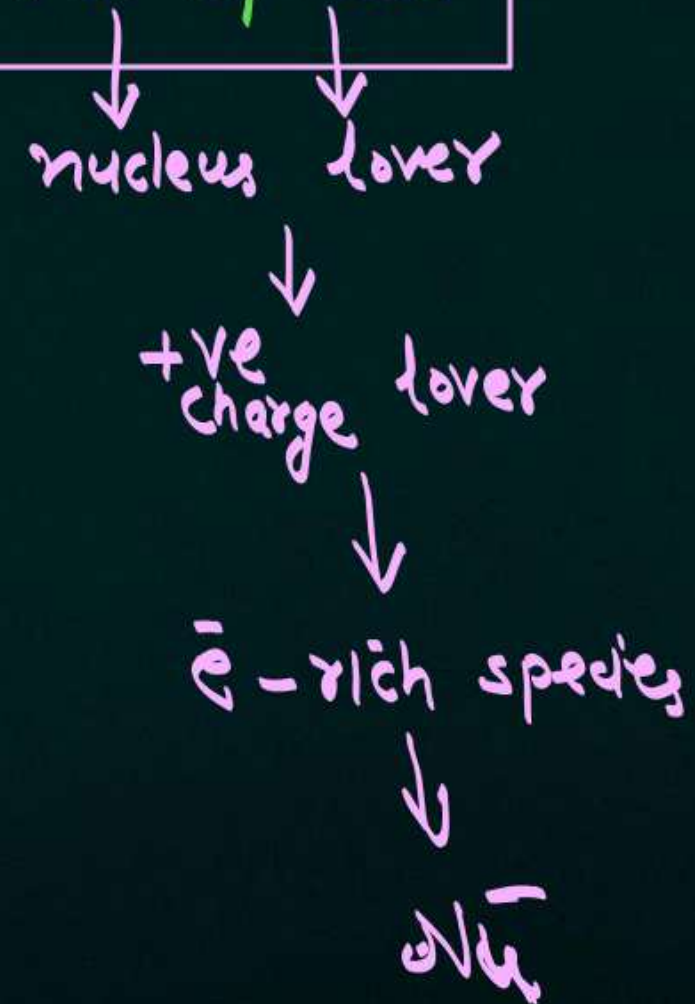


6 $\bar{e}$

Electrophile

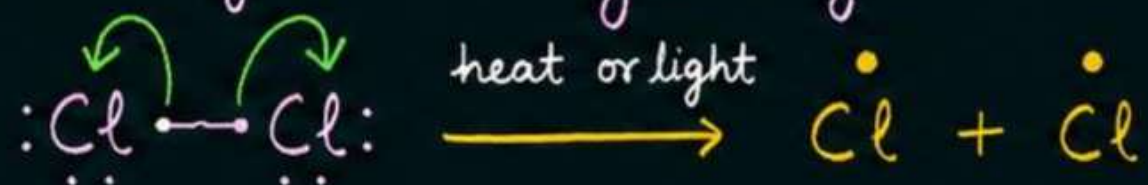


Nucleophile



FISSION OF COVALENT BOND.

1) HOMOLYTIC CLEAVAGE :→ If the covalent bond breaks in such a way that each atom takes away one $\bar{e}$ of shared pair (bond), it is called homolytic cleavage. e.g.



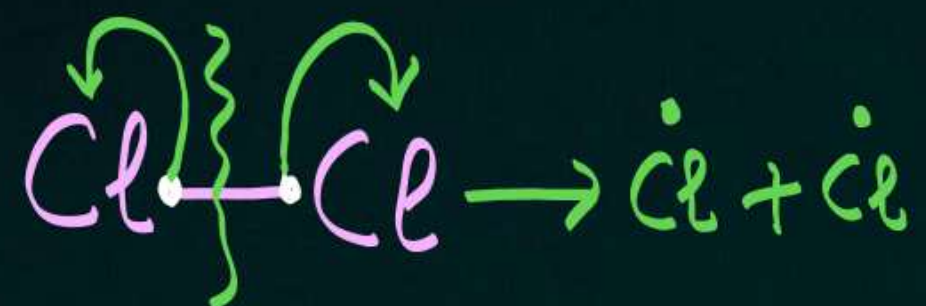
Free Radical

- These odd $\bar{e}$ species are called free Radicals.
- Homolytic cleavage is favoured by high T, UV light or peroxides.

2) HETEROLYTIC CLEAVAGE :→

If the covalent bond breaks in such a way that both the electrons of covalent bonds are taken away by one of the bonded atom (more electronegative atom), this type of cleavage is known as HETEROLYTIC CLEAVAGE. e.g.





Homol



Hetero

Q. Which of the following molecule is free radical.

- a) Cl^- b) Cl^+ c) $\text{Cl}^\bullet$ d) Cl_2

Guarantee
Never 100%.

FISSION OF COVALENT BOND



INTERMEDIATES $\rightarrow$

The short lived

highly reactive chemical species through which majority of organic reactions occurs are called reactive intermediates. Some important intermediates are discussed below.

1) CARBOCATION $\rightarrow$ Species having positive charge on carbon and carrying six electrons in their valence shell are called carbocation.

- It is formed by heterolytic cleavage.

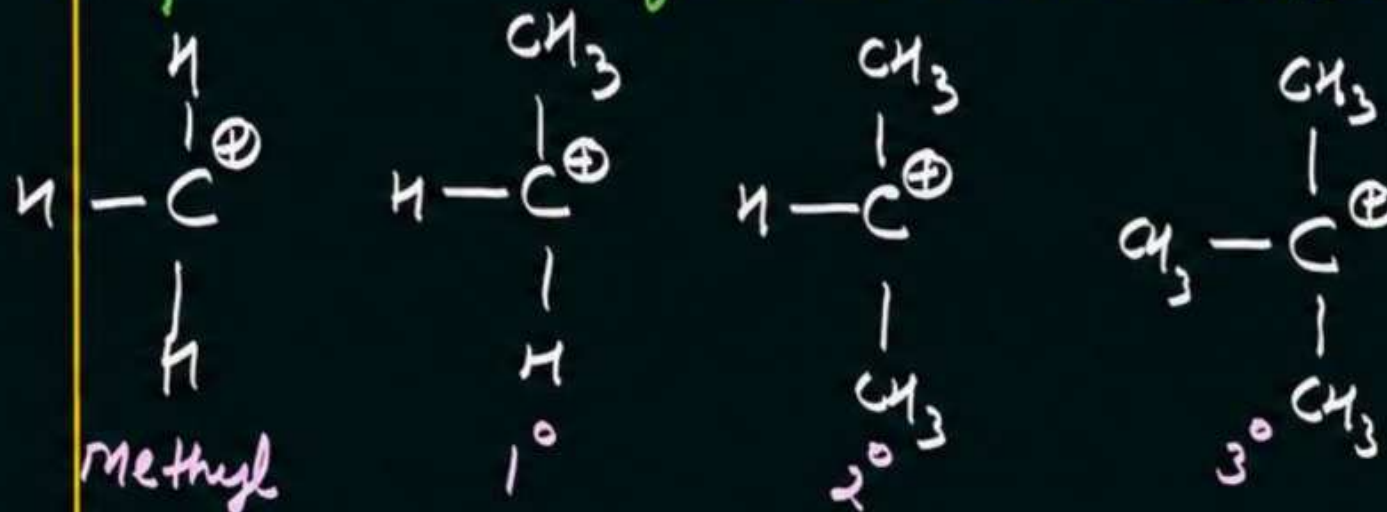


- It is $\bar{e}$ deficient species.



Classification $\rightarrow$

Classified as primary (1°), secondary (2°) & Tertiary (3°), depending upon the no. of carbon atoms attached. i.e.



STABILITY $\rightarrow$ Stability follows the order.



because. Carbocation is electron deficient species any $\bar{e}$ donating group stabilise the carbocation.

Tertiary has three $\bar{e}$ donating groups attached (via Inductive & hyperconjugative) i.e. why 3° is most stable.

(Resonance)
(-R)

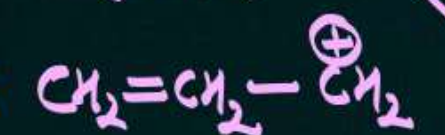


Benzyl

Carbocation

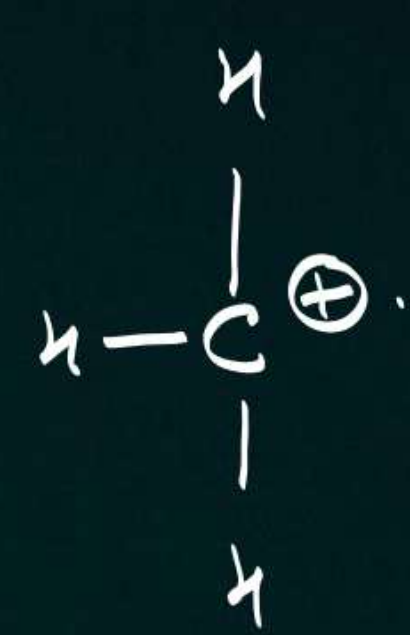
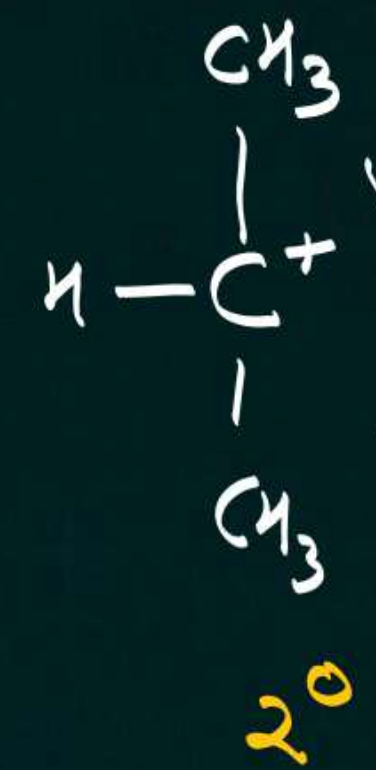
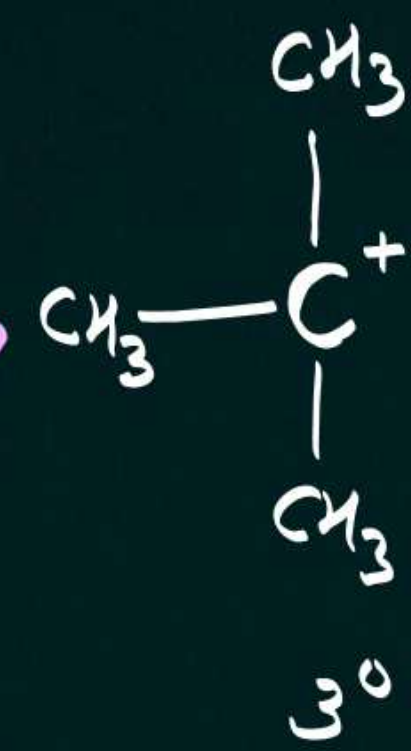
Reason $\Rightarrow$
Five resonating
structure
R $\uparrow$ stability $\uparrow$

Resonance
(-R)



Allyl
Carbocation

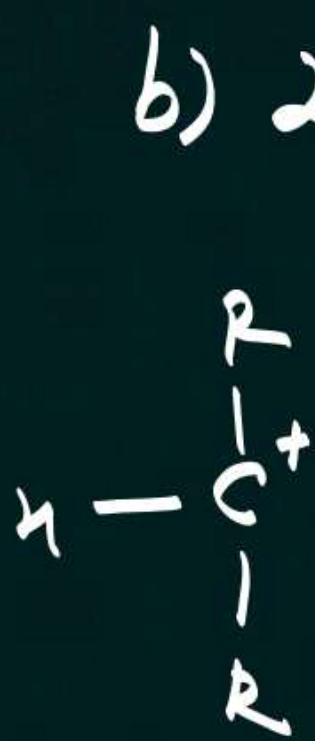
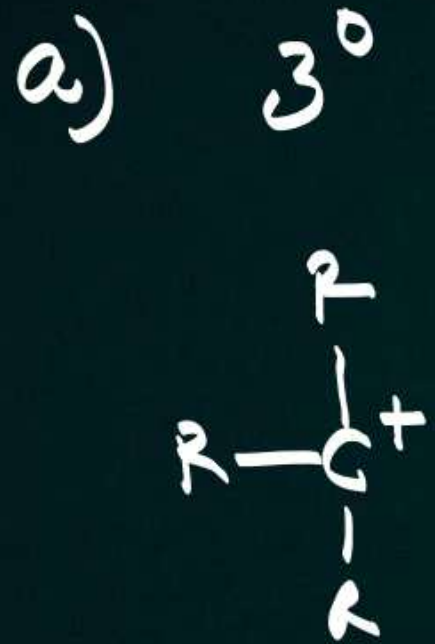
Reason =
Two Resonating
structure.



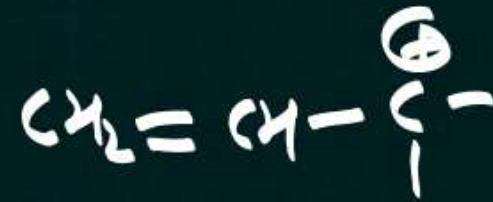
Alkyl carbocation

Order of strength $\Rightarrow$
Resonance $>$ Hyperconjugative $>$ Inductive

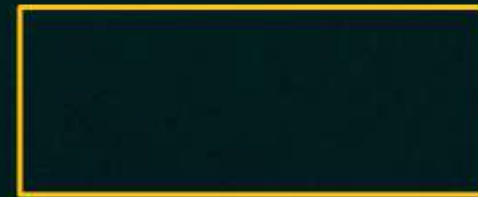
Q) Which is the following carbocation is most stable

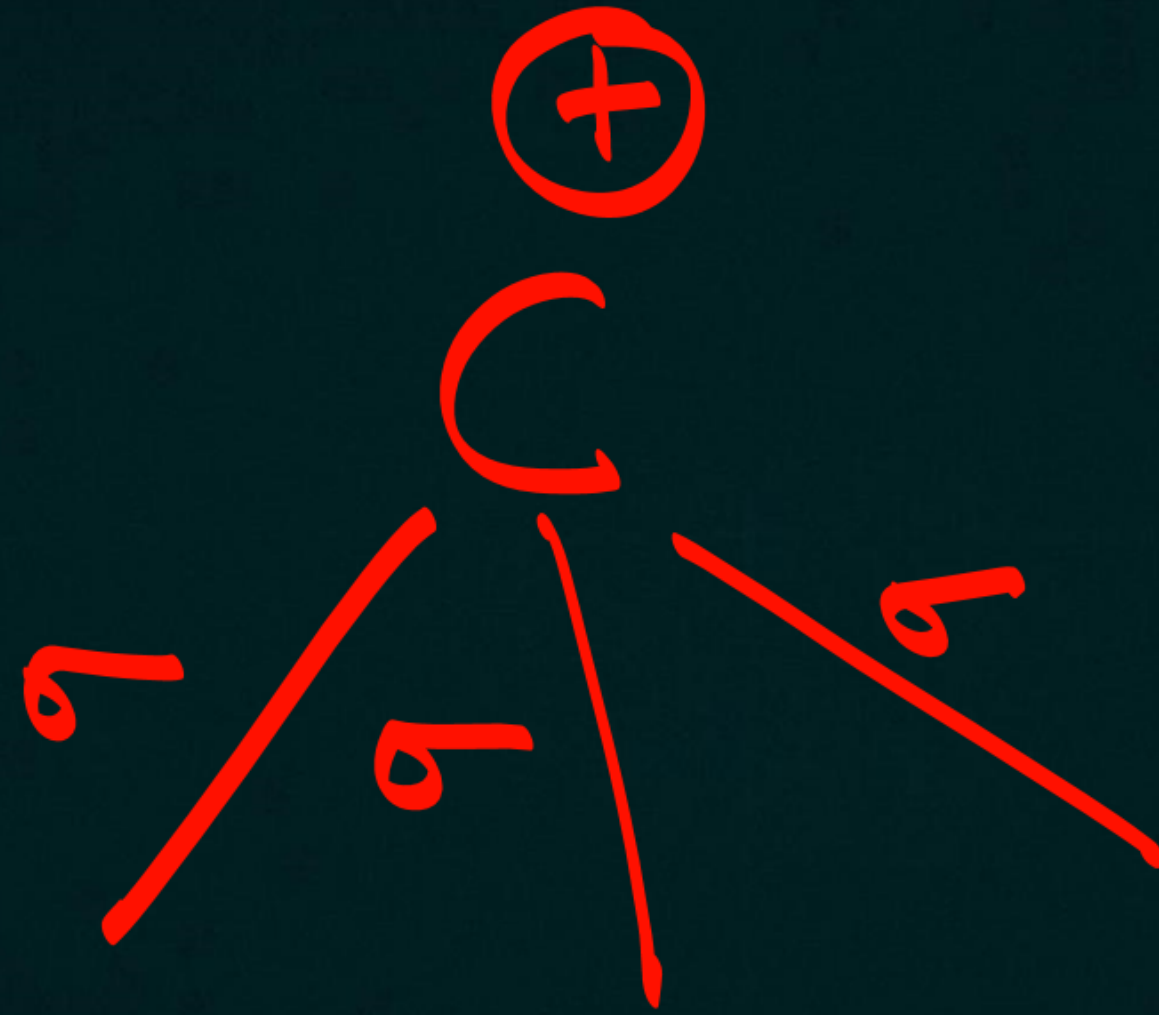


c) Allyl



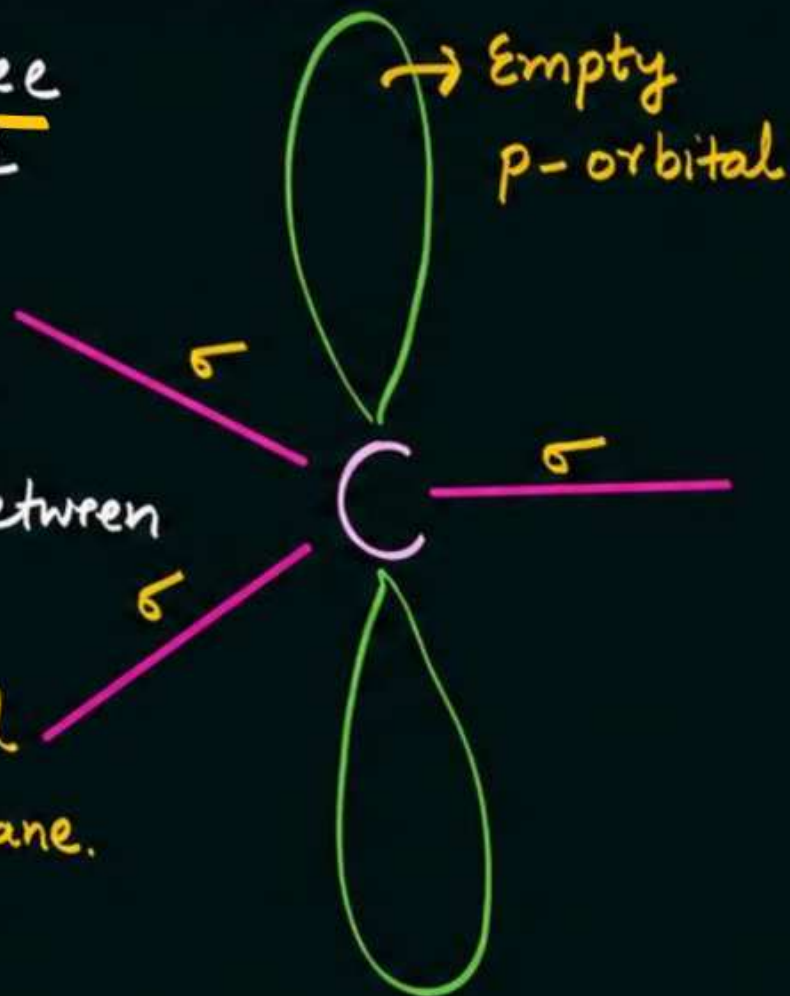
d) Benzyl.





Orbital structure of carbocation:→

- Carbon atom has three sigma bond & hence sp^2 hybridised
- It is trigonal planar with bond angle between σ -bonds is 120°
- One vacant-p-orbital is present $\perp$ to the plane.



$$TEP = \sigma + lp$$

2. Carbanion:→

Carboanion X

Chemical Species bearing a negative charge on carbon and having eight e^- 's in their valence shell are called carbanion.

- They are produced by heterolytic cleavage.

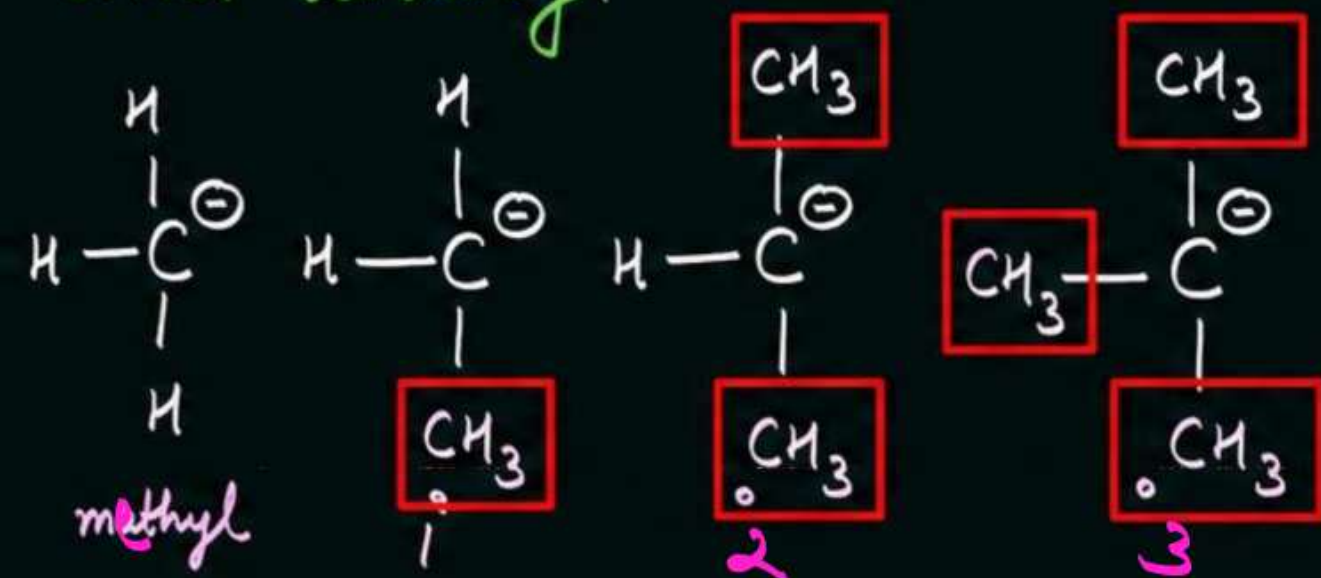


It is e^- rich species.



Classification of carbanion: →

It is also of 3 types i.e. primary, secondary and tertiary.



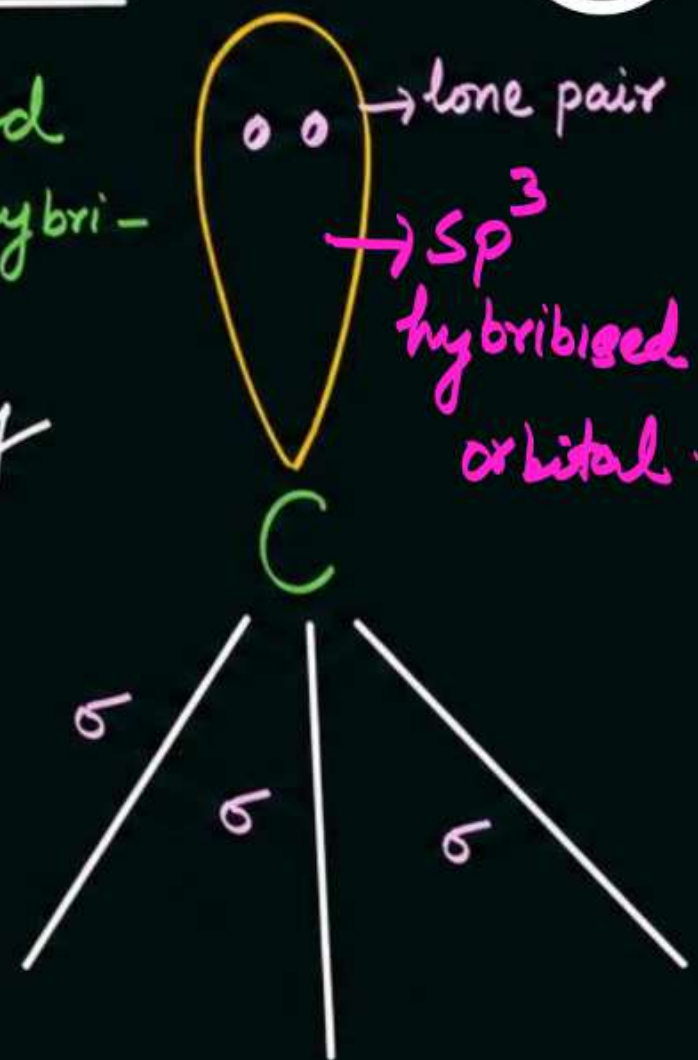
Order of Stability: → is methyl $> 1^\circ > 2^\circ > 3^\circ$

Because carbanion is $\bar{e}$ -rich species, therefore $\bar{e}$ withdrawing group will stabilise it. & $\bar{e}$ donating group will destabilise it.

3° carbanion has three $\bar{e}$ donating groups (via Inductive) therefore is least stable.

Orbital structure of Carbanion: →

- Carbon has 3- σ bonds and one lone pair, hence sp^3 hybridised.
- It has tetrahedral geometry and pyramidal shape.
- lone-pair is present in sp^3 hybridised orbital.



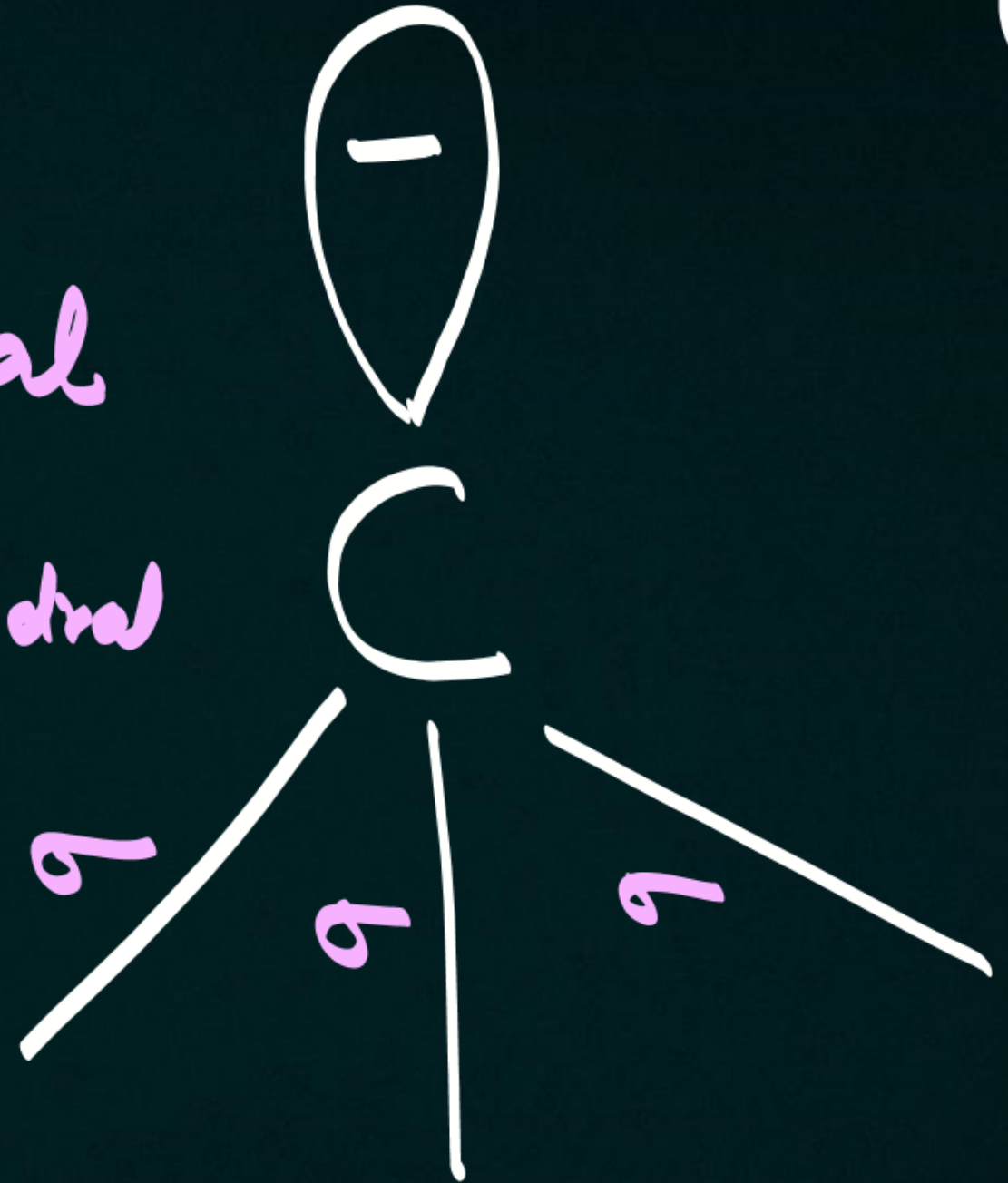
$$\begin{aligned} \text{TEP} &= \sigma + \text{lp} \\ &= 3 + 1 \\ &= 4 \rightarrow \underline{sp^3} \end{aligned}$$

sp^3 hyb.

Shape $\rightarrow$ Pyramidal
&

Geometry = Tetrahedral

lp = sp^3 hybridised
orbital.



$$\begin{aligned} \text{TEP} &= \sigma + \text{lp} \\ &= 3 + 1 \\ &= 4 \\ &= sp^3. \end{aligned}$$

Repeat Karwa
Dunga. Next
Class Me

Thank
You